

Dobot Magician E6 — quick guide

The arm is position controlled and does not feel you. It will not stop for your hand, the bench, or the phantom.

- Hold **L1** or nothing moves. Release it and the arm stops — that is your first reflex, not the emergency stop.
- Keep the **red emergency stop** on the base within reach. Press to stop, rotate to release.
- The **robot base is the centre** of the workspace: the arm reaches 450 mm in every direction, **front and back**. Nobody puts a hand inside it while it is running.

Start up

1. Power on the robot

Check the red emergency stop is released, switch on at the wall, then **tap** the power button on the base. Do not hold it.

 Wait **30-60 s** until the light is **steady blue**.

2. Connect the network

```
nmcli con up dobot-e6
```

3. Load the environment

```
cd "/home/yz22/Documents/code/Dobot arm"  
source setup_dobot.bash
```

4. Send the arm home

```
python3 src/dobot_e6_hw/scripts/go_home.py --speed 5
```

 Wait until it prints **Arm at home pose**.

Probe **not** fitted? Add `--no-probe` here and `probe:=false` in step 5.

5. Start the program

```
ros2 launch dobot_e6_hw real_hw.launch.py robot_ip:=192.168.5.1
```

 Wait about **15 s** until the light is **flashing green slowly**, then **25 s more** — until **Calibration complete** appears — before touching the controller. About **40 s** in total.

Stop the program: press `Ctrl+C` in the same terminal. The light returns to steady blue.

Turn off the robot: after stopping the program, **support the probe with one hand** and **hold** the power button until the light **flashes red**, then release. The light goes off.

Restart: press `Ctrl+C`, then repeat **steps 4-5**. If the robot was turned off or the emergency stop was pressed, repeat from **step 1**.

Yellow light (collision detected): `Ctrl+C`, check nothing is touching the arm, then clear it without moving the arm and repeat steps 4-5:

```
python3 src/dobot_e6_hw/scripts/clear_alarm.py
```

Joystick



HOME: hold L1 + L2 + R2 for 1 s, then release L1 and press it again to drive

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